

Samaranjan Manjhi

+91 9082650120 | manjhisamaranjan@gmail.com | [LinkedIn](#) | [Portfolio](#) | Kandivali, Mumbai

SUMMARY

Software engineer specializing in system-level programming and security solutions with 2.5+ years of experience in C/C++ development. Expertise in building cross-platform monitoring tools, file integrity systems, and kernel-level security solutions for macOS and Linux using Qt framework and security libraries.

TECHNICAL SKILLS

Languages & Frameworks: C/C++, Qt

Operating Systems & Tools: Linux, macOS, Vim, Xcode, Git, SQLite

Libraries: OpenSSL, cURL, Leptonica, Tesseract, Poppler, libmagic, libarchive, Endpoint Security, libcups

PROFESSIONAL EXPERIENCE

Programmer

August 2023 – Present

MicroWorld Software Services Pvt Ltd

Andheri, Mumbai

- Engineered macOS process monitoring, file integrity tools, and cross-platform cloud backup solution with real-time tracking, automated monitoring, content scanning, and alerting to enhance system security and ensure data integrity.
- Implemented Linux kernel module with custom system call hooks for ransomware protection and developed Qt-based vulnerability scanning application with SQLite database for comprehensive security assessment.

KEY PROJECTS

Printer Control & Content Filter | C/C++, Xcode, Endpoint Security, libcups

Oct 2025 – Jan 2026

- Developed macOS printer control system enforcing application-based whitelisting/blacklisting and content scanning for sensitive data during print operations, blocking unauthorized prints based on process and content check status, with comprehensive logging using Endpoint Security framework and libcups.

ProcMon - Process Monitoring Tool | C/C++, libmagic, bsm

July 2025 - Sept 2025

- Built lightweight macOS process monitoring solution capturing new executions (scp, rsync, vim) with command-line arguments and process hierarchies using native system calls, eliminating dependencies on Endpoint Security framework.

File Integrity Monitoring System | C/C++, OpenSSL, SQLite3

June 2025 - Aug 2025

- Created macOS FIM system processing configuration-based watch lists, maintaining metadata (permissions, MD5, size, timestamps) in SQLite, continuously detecting real-time modifications with alerting and audit logging.

cBackup - Cloud Backup & Monitoring Tool | C/C++, OpenSSL, cURL

Feb 2025 - May 2025

- Architected cross-platform backup solution for macOS/Linux monitoring file transfers (USB, scp, rsync, curl, email), automatically creating local backups and synchronizing across cloud services with 10-minute intervals to prevent data loss.

Content Scanner | C/C++, Leptonica, Tesseract, Poppler

Sept 2024 - Dec 2024

- Developed data scanning solution detecting confidential information (PAN, emails) across 12+ file formats (PDF, images, XML, DOCX, archives) using regex patterns, available as shared library and CLI tool.

System Call Interception Module | C, Linux Kernel, Hooking

Mar 2024 - Aug 2024

- Developed Linux kernel module implementing system call interception with custom hooks for monitoring file operations by extension (creation, renaming, moving), enhancing security against ransomware attacks.

Vulnerability Checker Application | C/C++, Qt, Boost, cURL, OpenSSL, cJSON

Oct 2023 - Jan 2024

- Built vulnerability assessment application integrating Qt frontend with C++ backend, utilizing cURL for automated ZIP downloads, XML parsing, and SQLite for efficient vulnerability data storage and scanning workflows.

EDUCATION

Bachelor of Engineering in Computer Engineering

2019 – 2023

Shree LR Tiwari College Of Engineering, Mumbai University

CGPA: 8.56/10